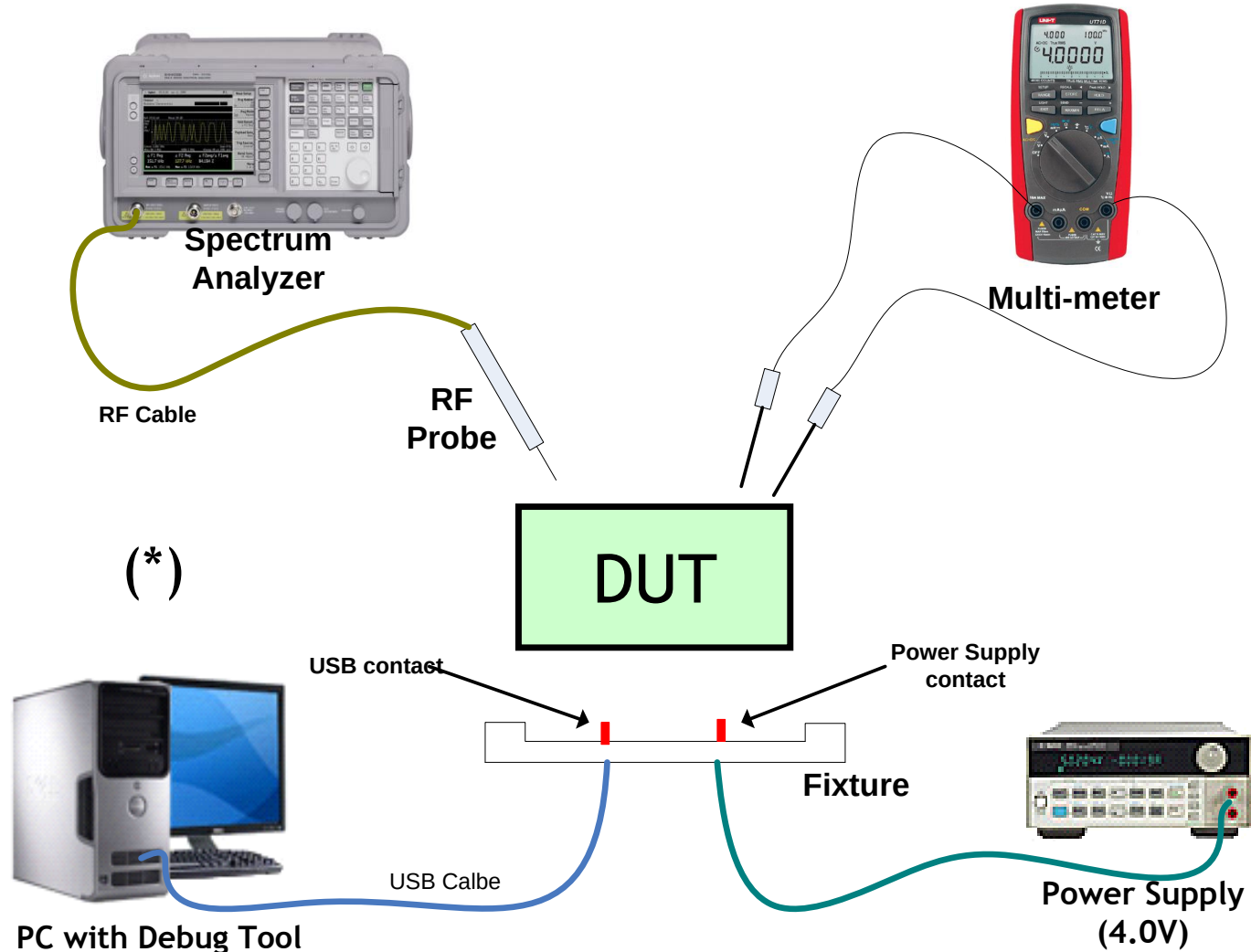


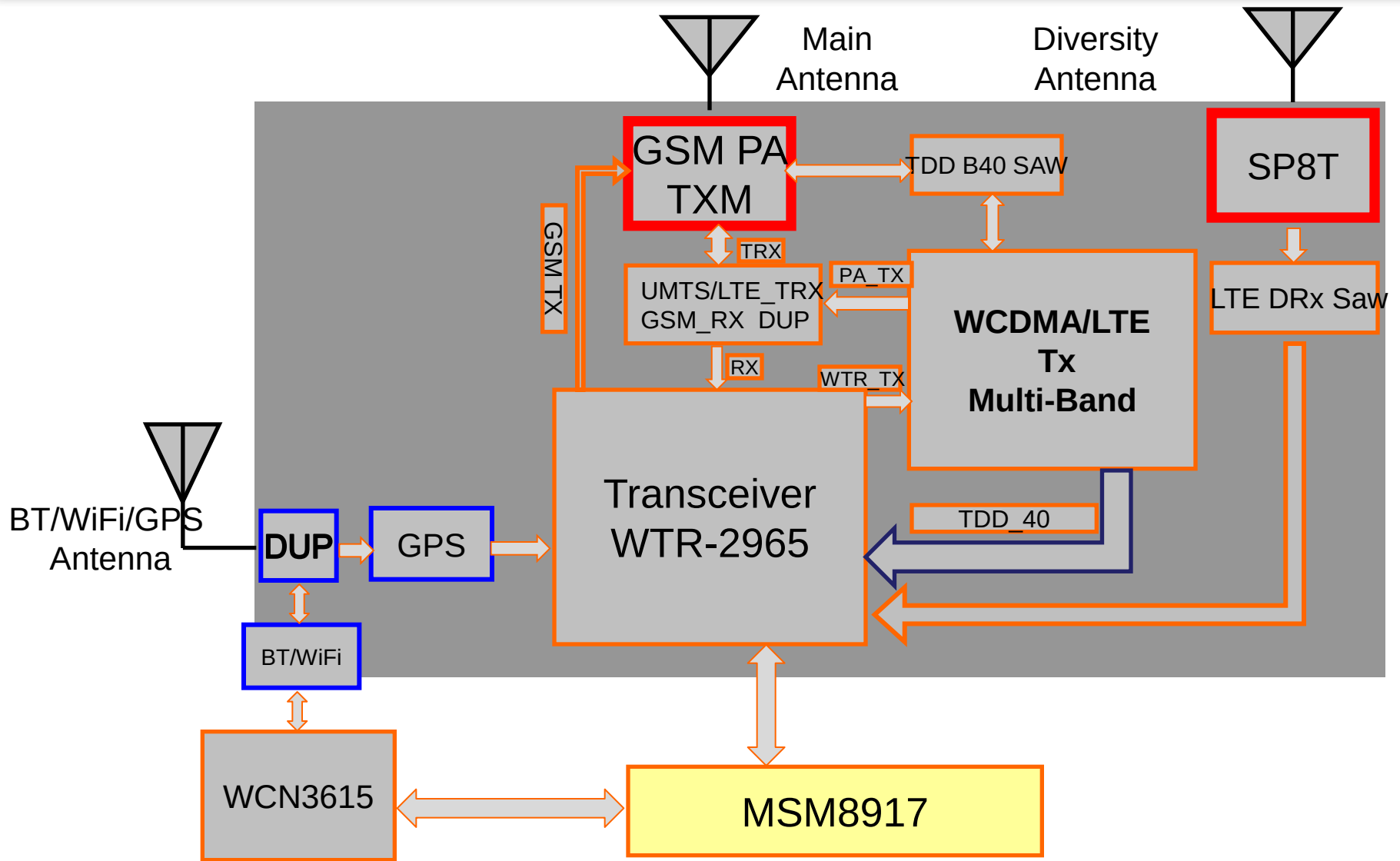
ASUS ZA550KL

RF Hardware Trouble Shooting Guide

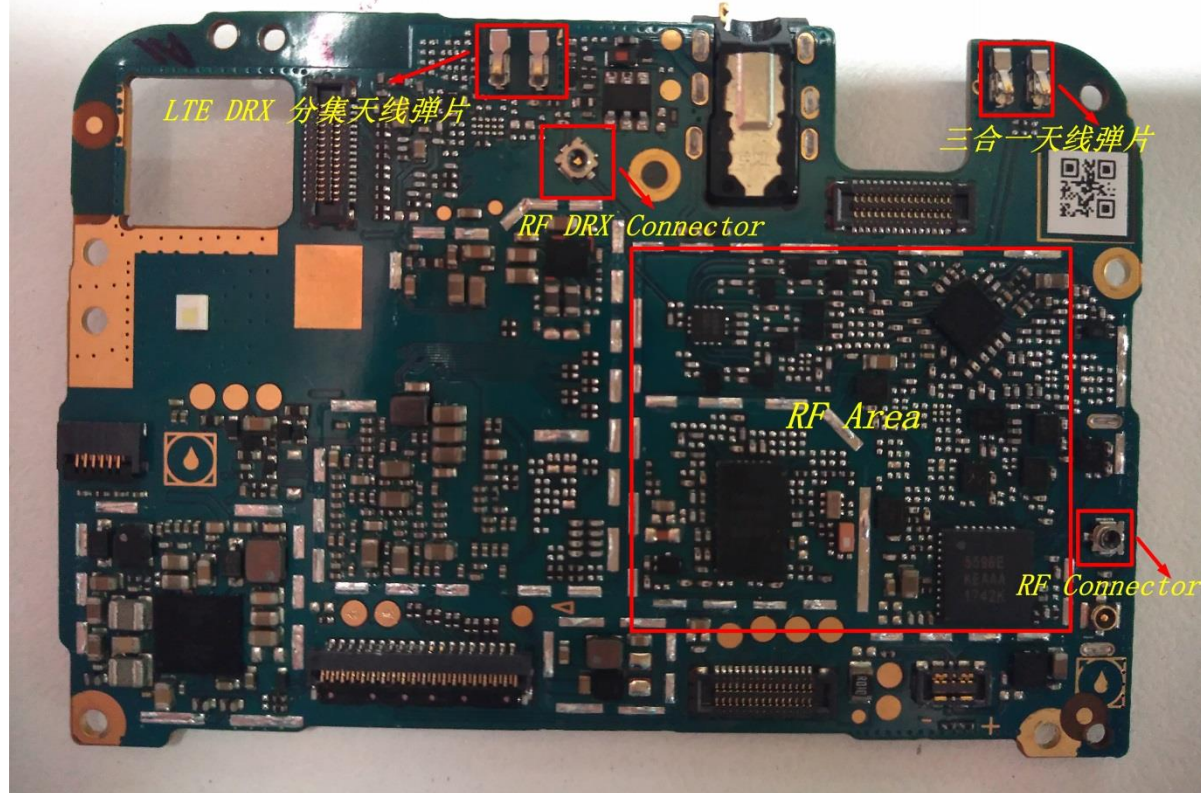
Repair and Test Environment



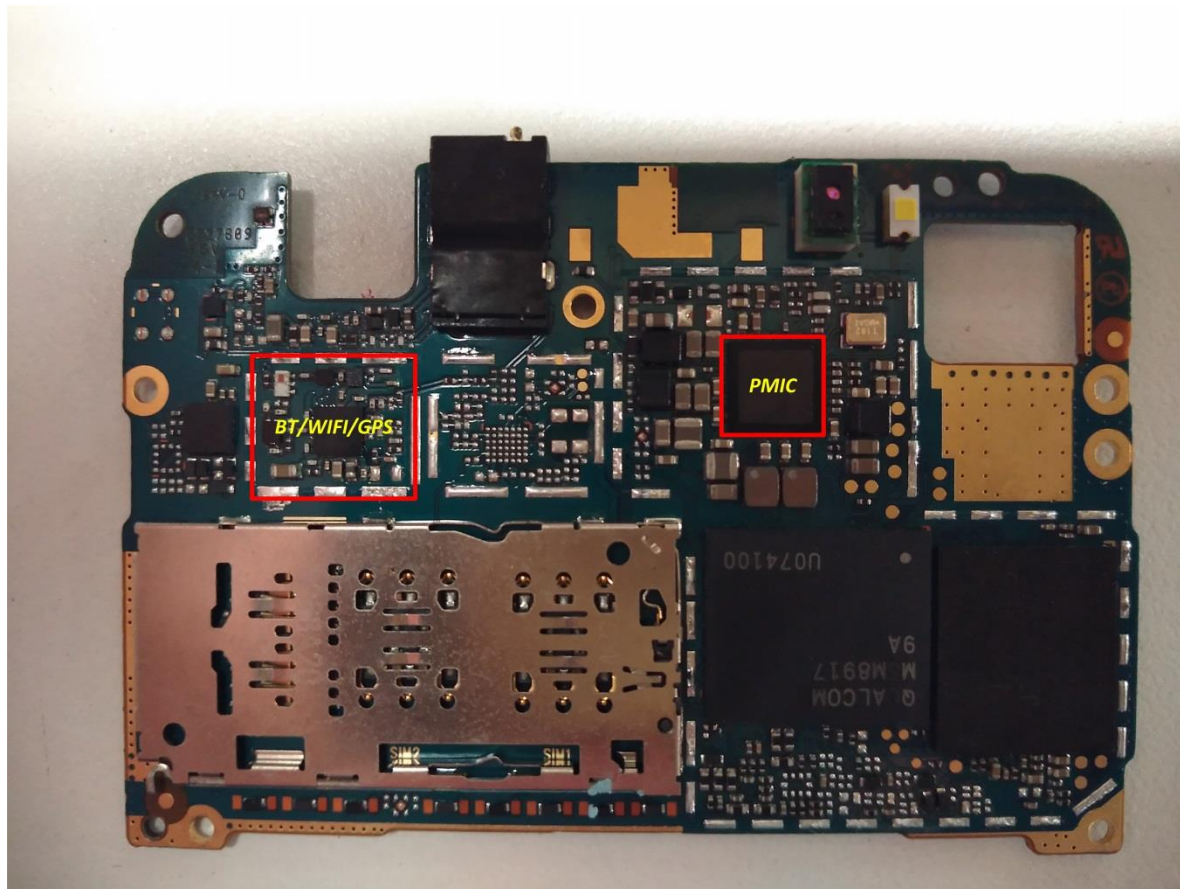
RF Block Diagram



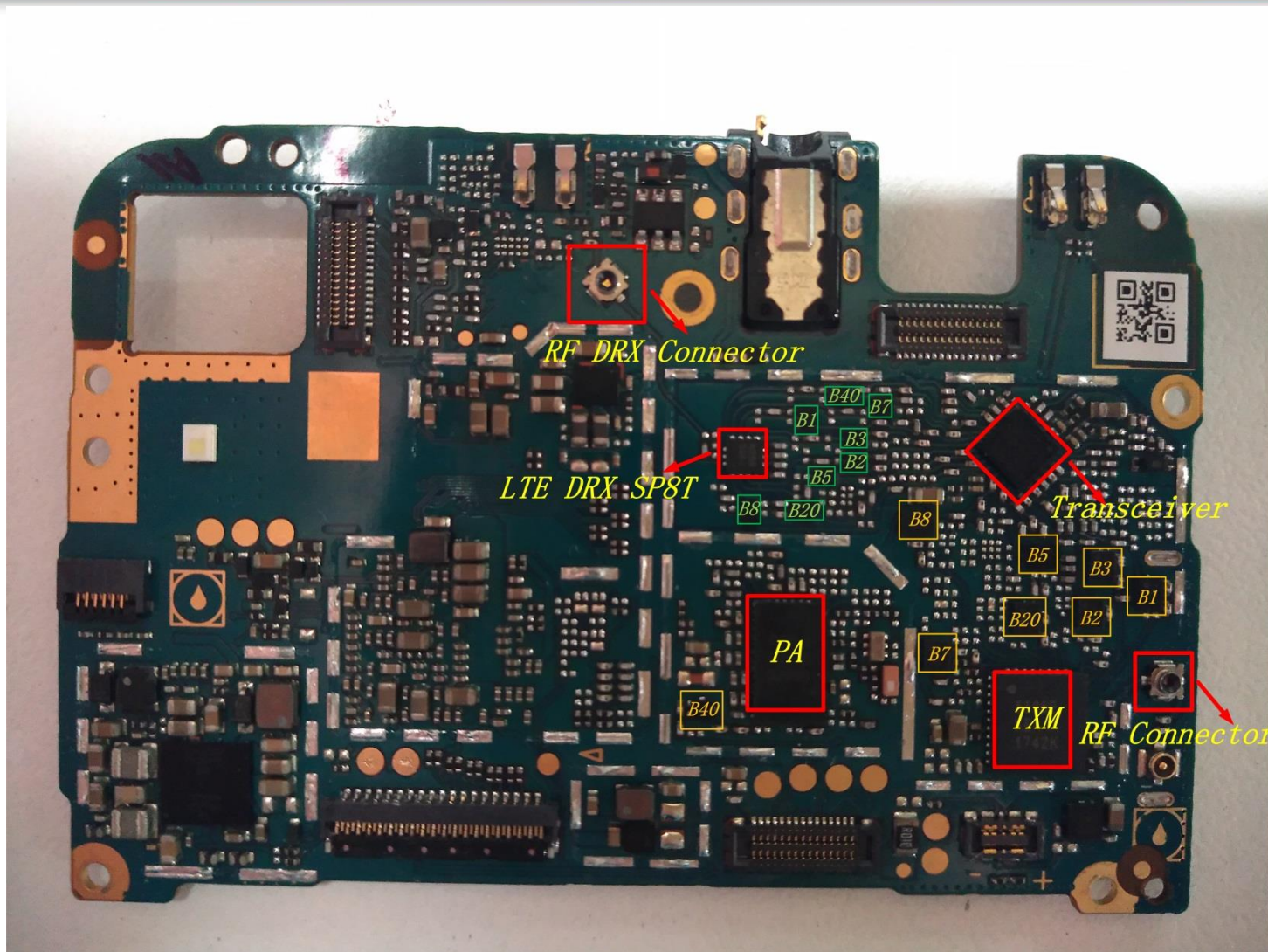
RF areas



RF areas



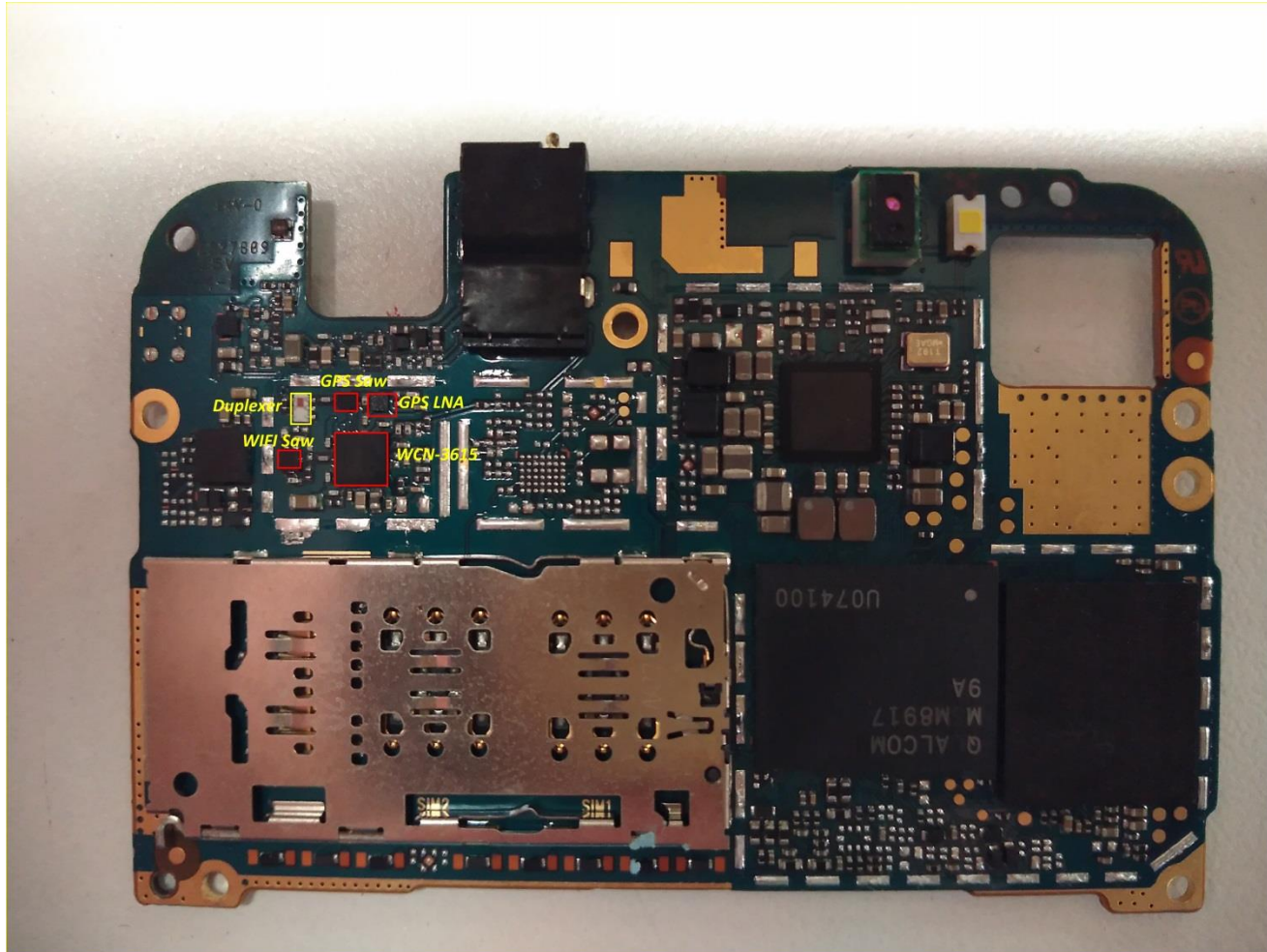
RF Key Component (Bottom)



Saw

Duplexer

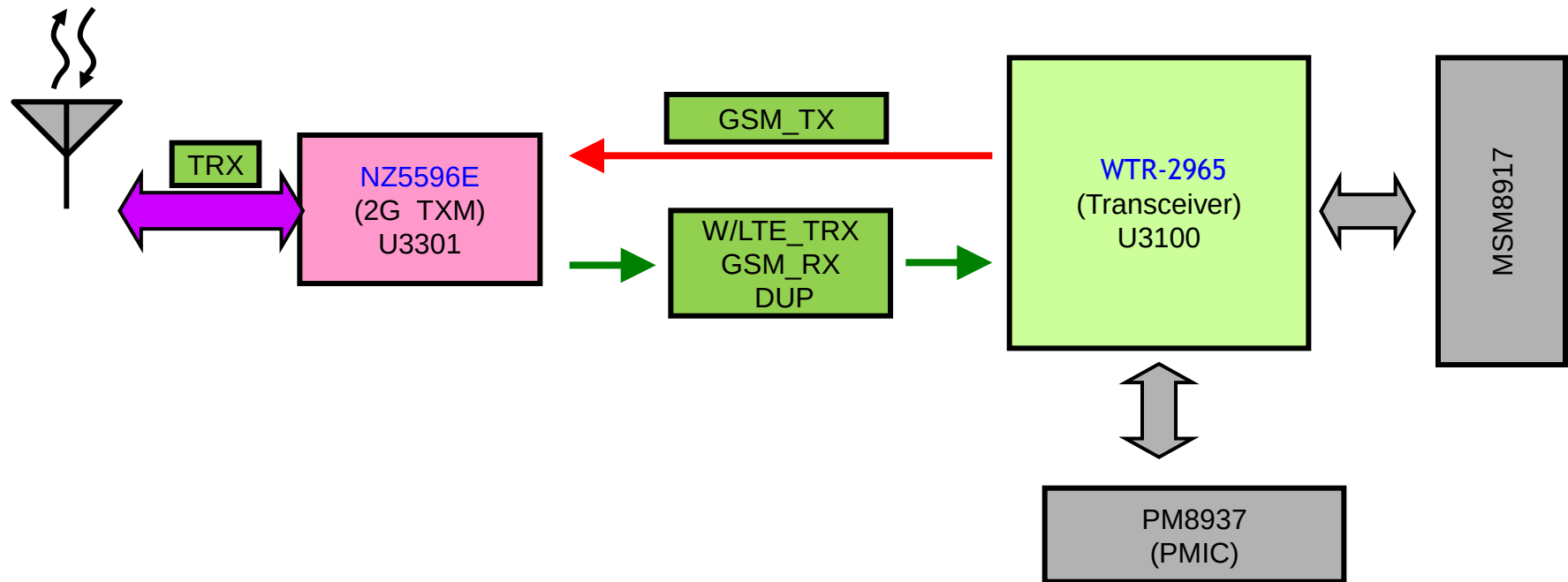
RF Key Component (Top)



Duplexer

SAW

GSM Block Diagram



Antenna Switch(U3301) –NZ5596E

Register 0	Description	Default	Notes
[7]	Power Control Selector	0	(set to 1 to select this Vramp GMSK Power Control register map)
[6:5]	Reserved	00	
[4:0]	Mode Control	00000	0x00 = Standby 0x01 = TRx5 0x02 = TRx4 0x03 = TRx3 0x04 = TRx9 0x05 = TRx6 0x08 = TRx10 0x09 = TRx7 0x0A = LB GMSK Tx 0x0B = LB EDGE Tx 0x0D = TRx8 0x0E = HB GMSK Tx 0x0F = HB EDGE/TD Tx 0x10 = TRx1 0x11 = TRx13 0x14 = TRx11 0x18 = TRx2 0x19 = TRx14 0x1C = TRx12 Other = Isolation

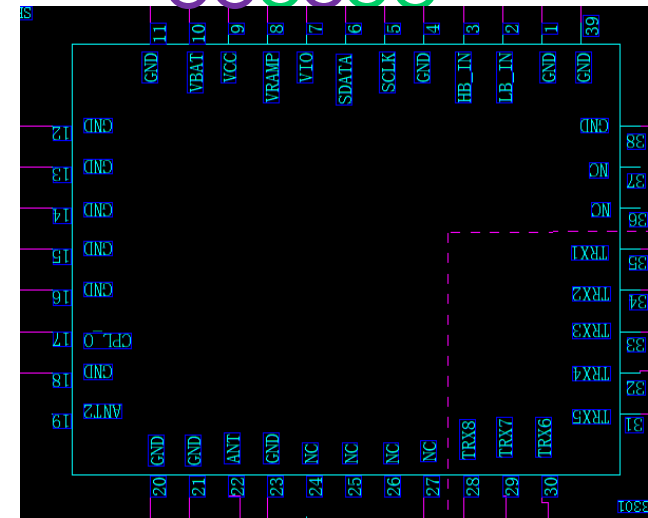
 Control pin
  Power pin

1 2 3 4 5 6



Test point locations

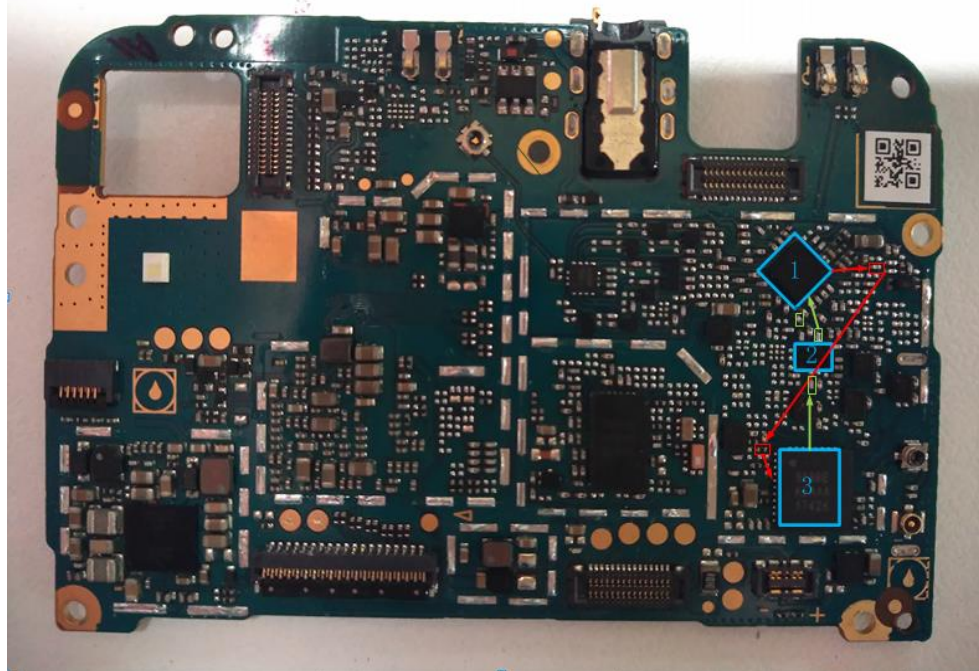
1. VBAT: 3.8V/0V
2. VCC: 3.8V/0V
3. VRAMP(APC1)
4. VIO(LTE_VMIPI): 1.8V/0V
5. SDATA(MIPI1_SDATA)
6. SCLK(MIPI1_SCLK)



GSM850

— Tx path
— Rx path

1. U3100: Transceiver
WTR-2965
2. U3405: Band5 Duplexer
3. U3301: TXM
NZ5596E



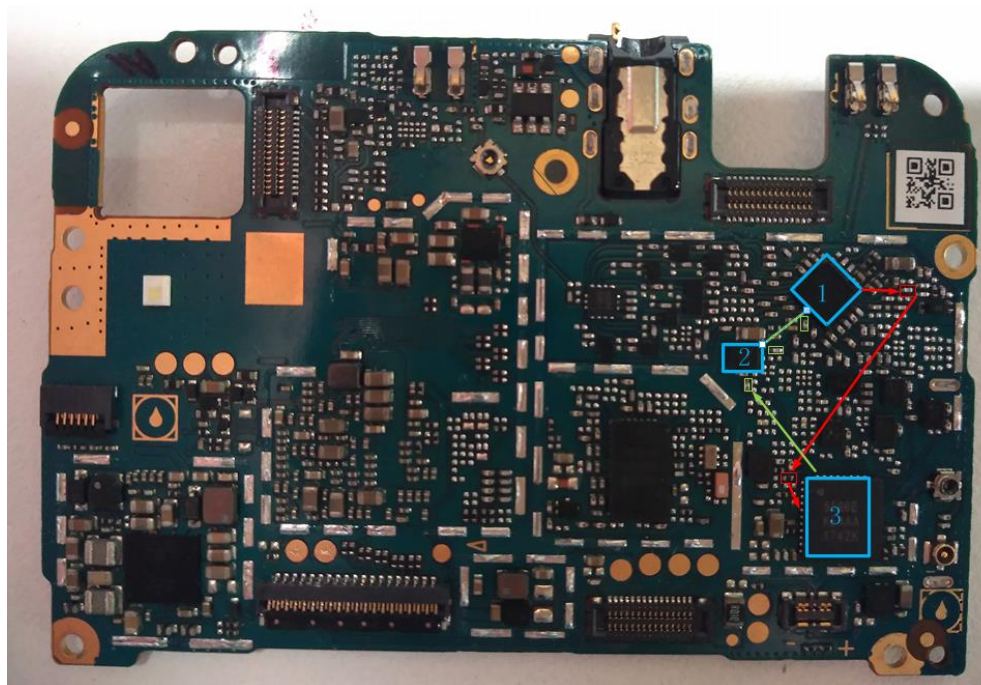
Rx Serial comp.
 Tx Serial comp.
 Serial IC.

RF (Bottom)

EGSM

— Tx path
— Rx path

1. U3100: Transceiver
WTR-2965
2. U3408: Band8 Duplexer
3. U3301: TXM
NZ5596E



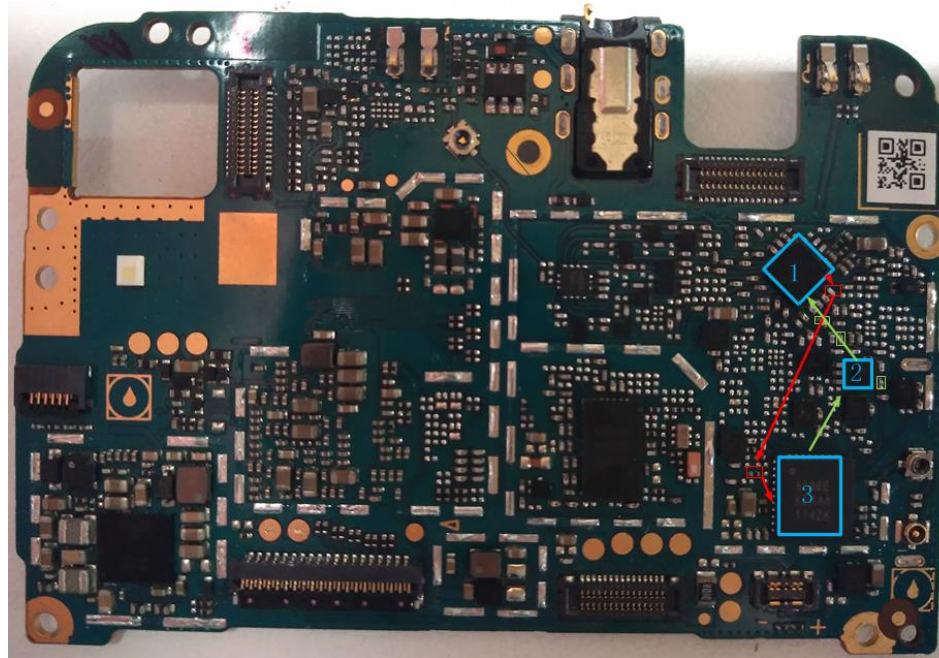
Rx Serial comp.
 Tx Serial comp.
 Serial IC.

RF (Bottom)

— Tx path
— Rx path

1. U3100: Transceiver
WTR-2965
2. U3403: B3 Duplexer
3. U3301: TXM
NZ5596E

□ Rx Serial comp.
□ Tx Serial comp.
□ Serial IC.

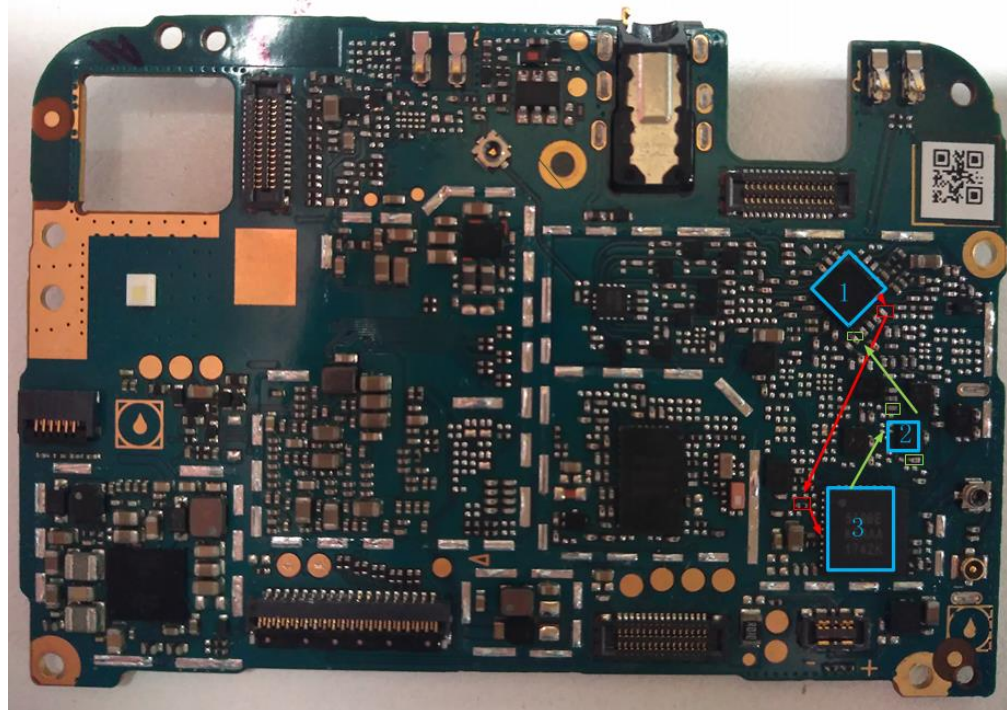


RF (Bottom)

— Tx path
— Rx path

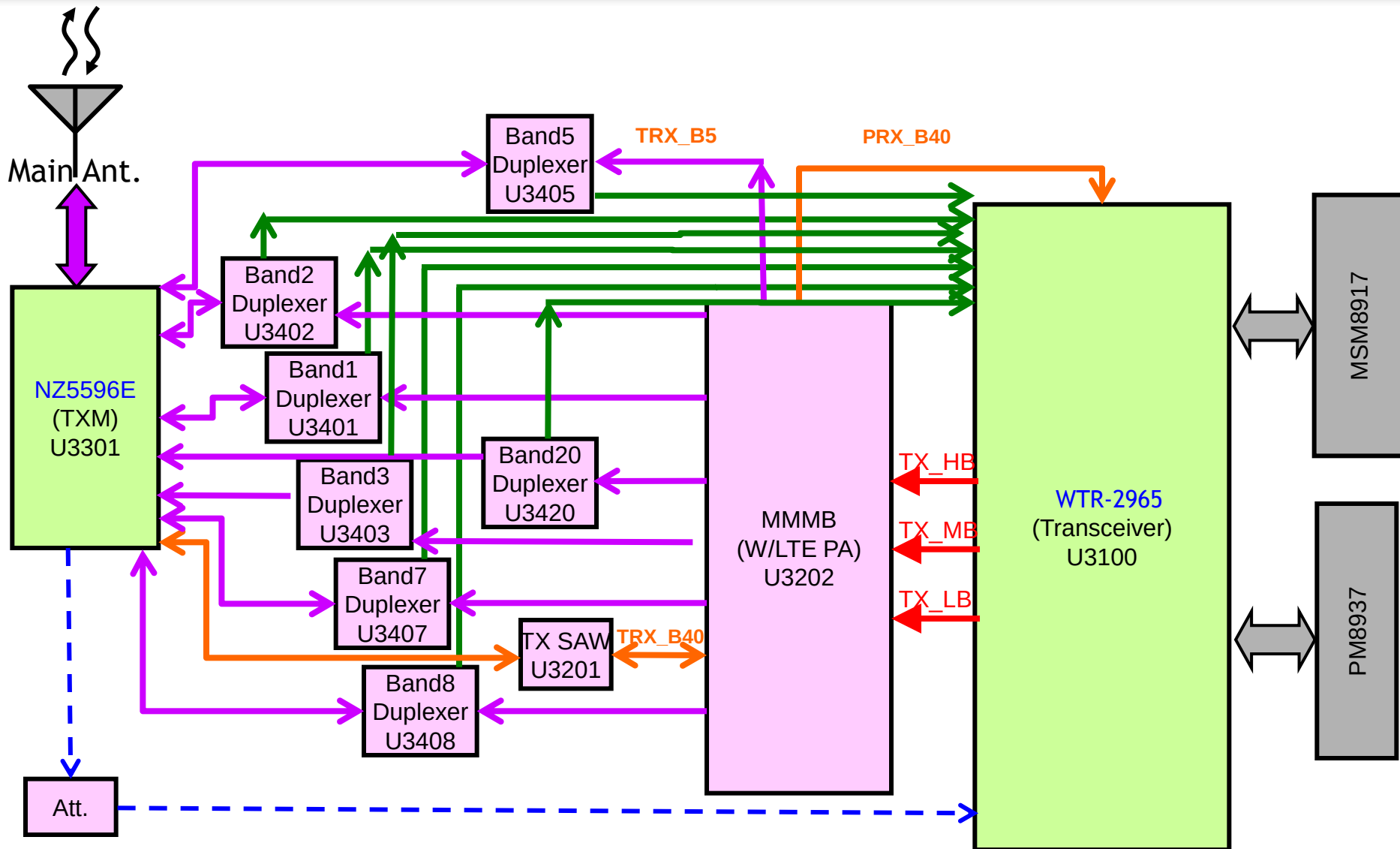
1. U3100: Transceiver
WTR-2965
2. U3403: Band2 Duplexer
3. U3301: TXM
NZ5596E

□ Rx Serial comp.
□ Tx Serial comp.
□ Serial IC.



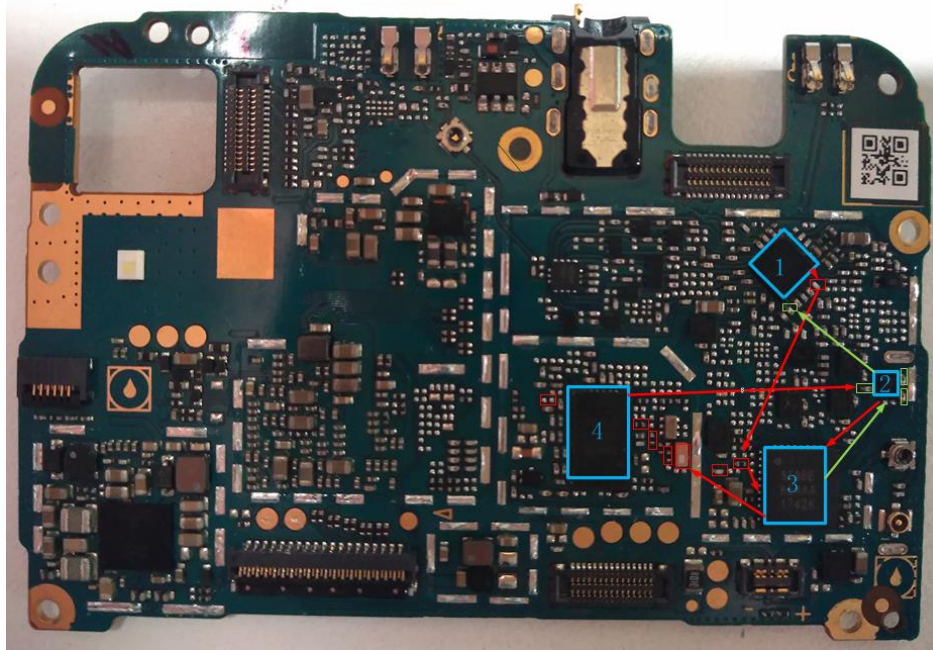
RF (Bottom)

WCDMA/LTE Main Ant Block Diagram



— Tx path
— Rx path

1. U3100: Transceiver
WTR-2965
2. U3401: Band1 Duplexer
3. U3301: TXM
NZ5596E
4. U3202: MMBB
NZ5627E



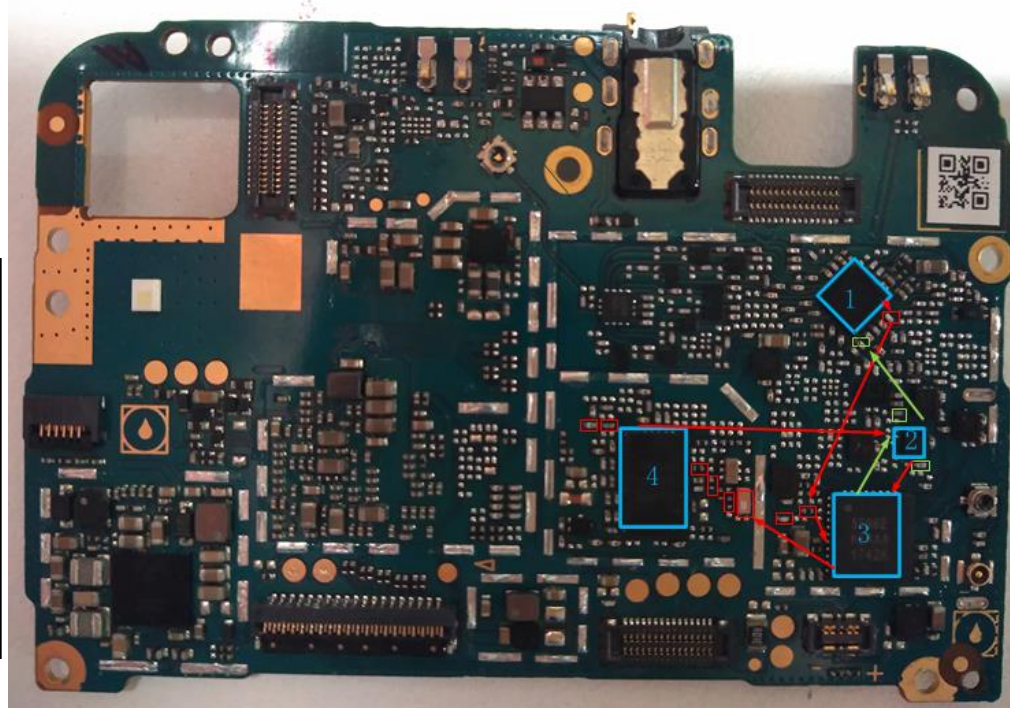
Rx comp.
 Tx comp.
 Serial IC.

RF (Bottom)

— Tx path
— Rx path

□ Rx comp.
□ Tx comp.
□ Serial IC.

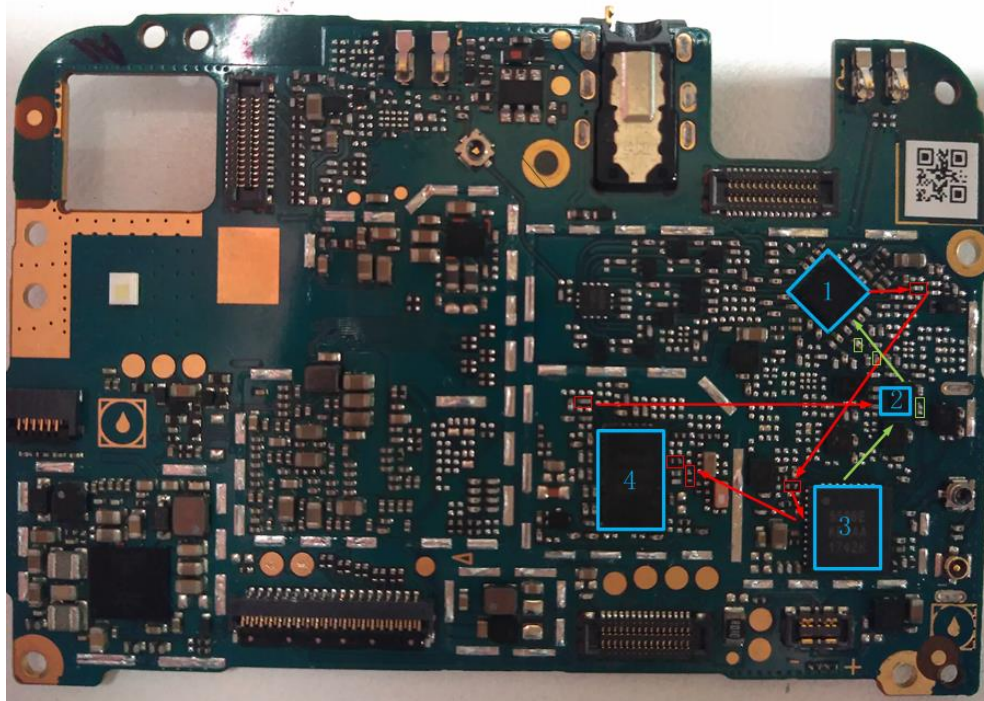
1. U3100: Transceiver
WTR-2965
2. U3402 : Band2 Duplexer
3. U3301: TXM
NZ5596E
4. U3202: MMBB
NZ5627E



RF (Bottom)

— Tx path
— Rx path

1. U3100: Transceiver
WTR-2965
2. U3403 : Band3 Duplexer
3. U3301: TXM
NZ5596E
4. U3201: MMBB
NZ5627E

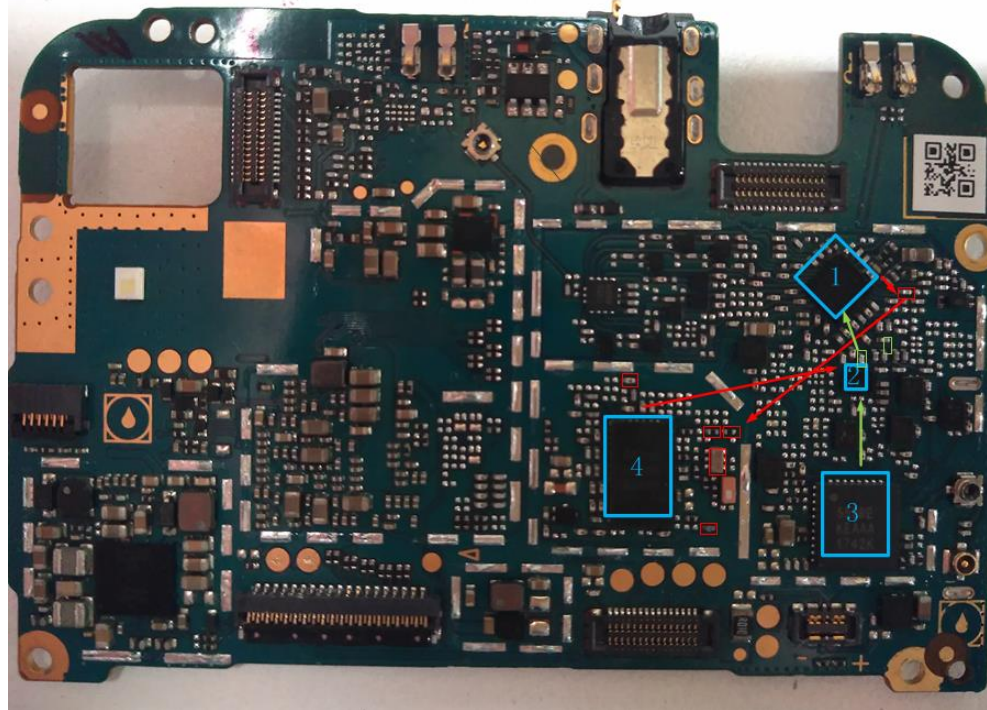


□ Rx comp.
□ Tx comp.
□ Serial IC.

RF (Bottom)

— Tx path
— Rx path

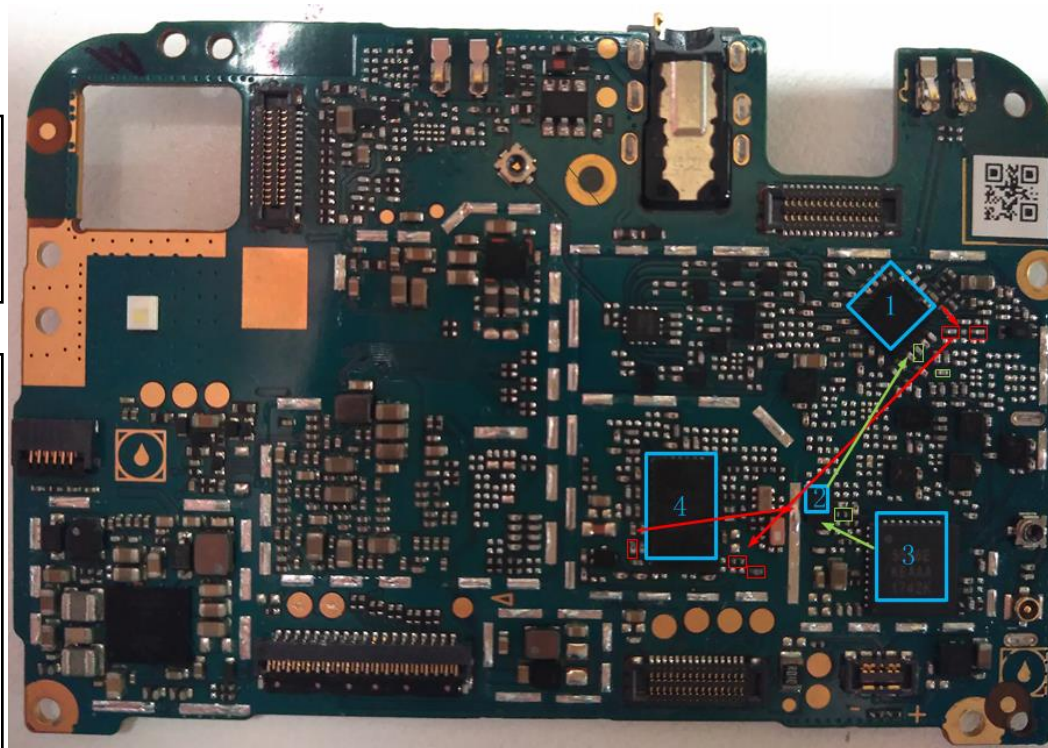
□ Rx comp.
□ Tx comp.
□ Serial IC.



RF (Bottom)

— Tx path
— Rx path

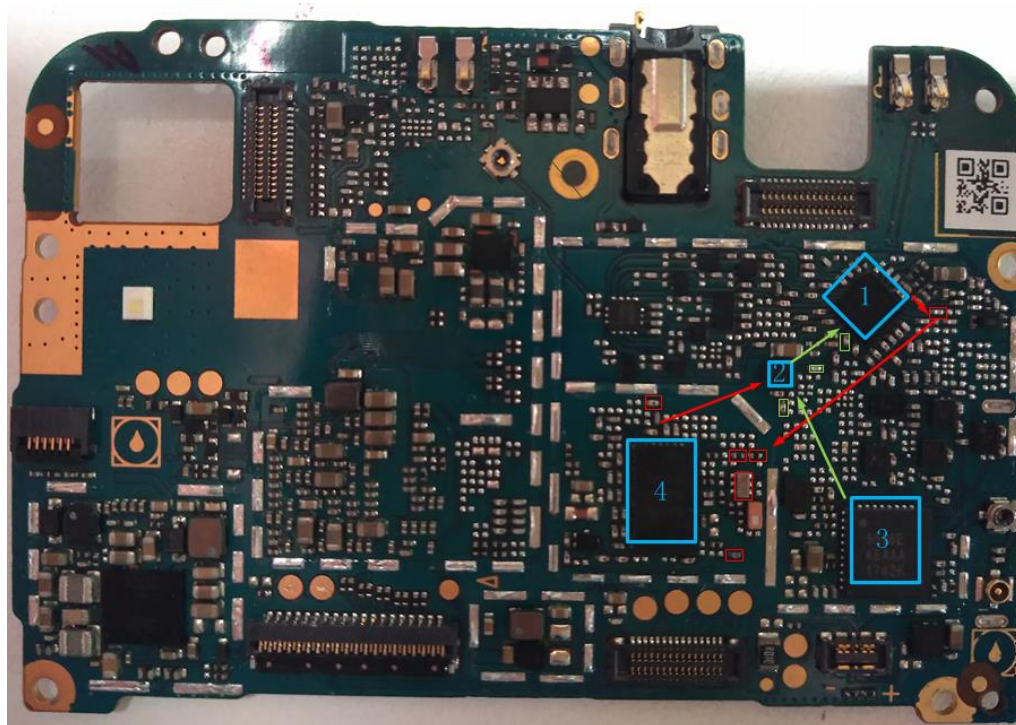
□ Rx comp.
□ Tx comp.
□ Serial IC.



RF (Bottom)

— Tx path
— Rx path

1. U3100: Transceiver
WTR-2965
2. U3408 : Band8 Duplexer
3. U3316: TXM
NZ5596E
4. U3201: MMBB
NZ5627E



□ Rx comp.
□ Tx comp.
□ Serial IC.

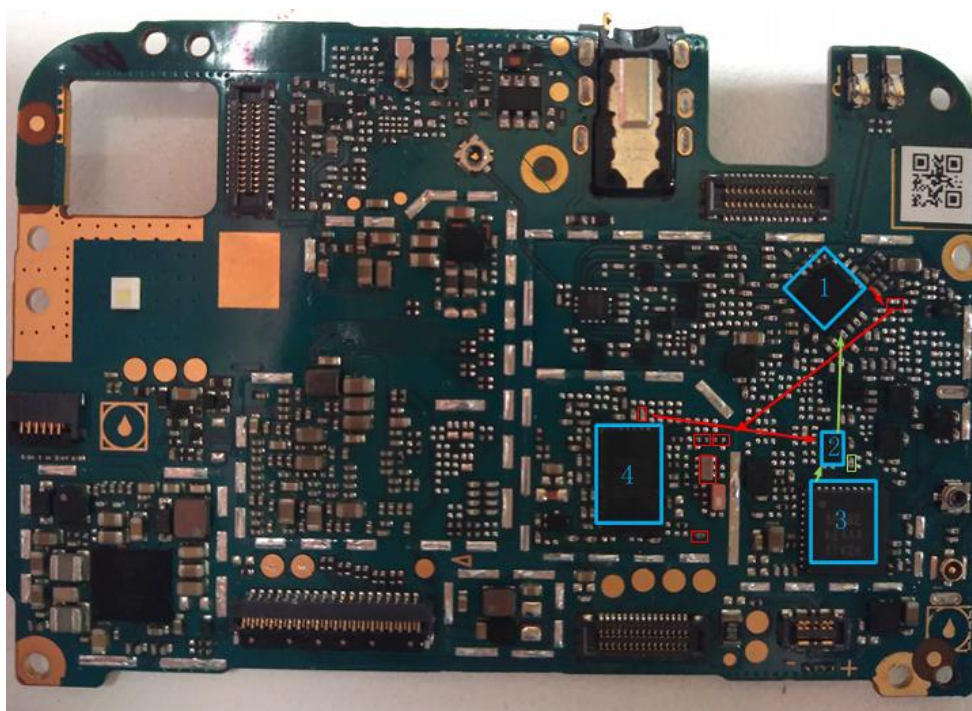
RF (Bottom)

B20

— Tx path
— Rx path

1. U3100: Transceiver
WTR-2965
2. U3420 : Band20 Duplexer
3. U3316: TXM
NZ5596E
4. U3201: MMBB
NZ5627E

Rx comp.
Tx comp.
Serial IC.

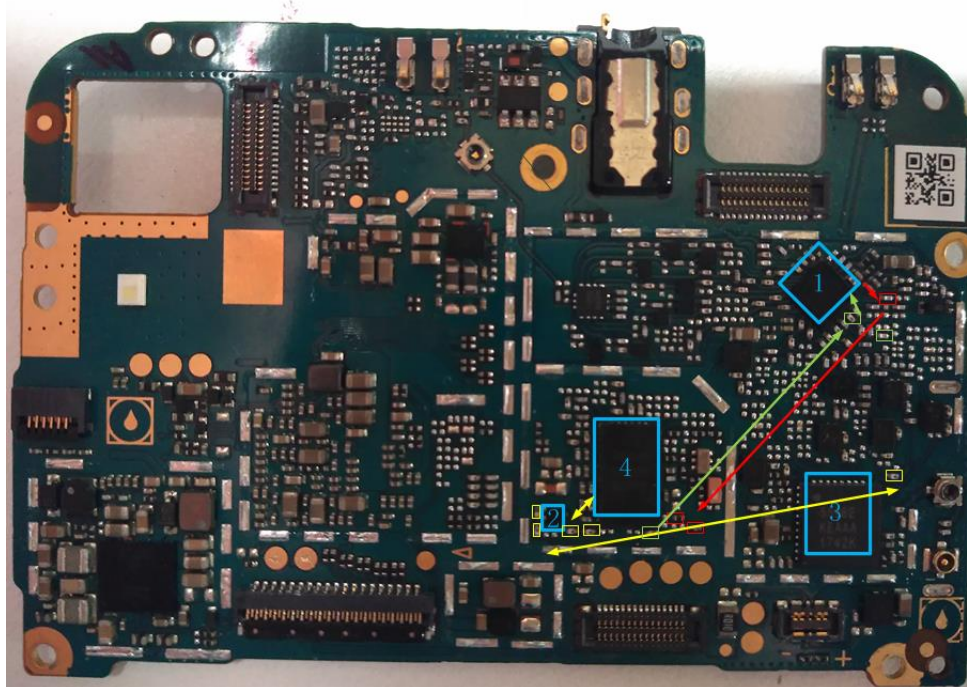


RF (Bottom)

B40

— Tx path
— Rx path

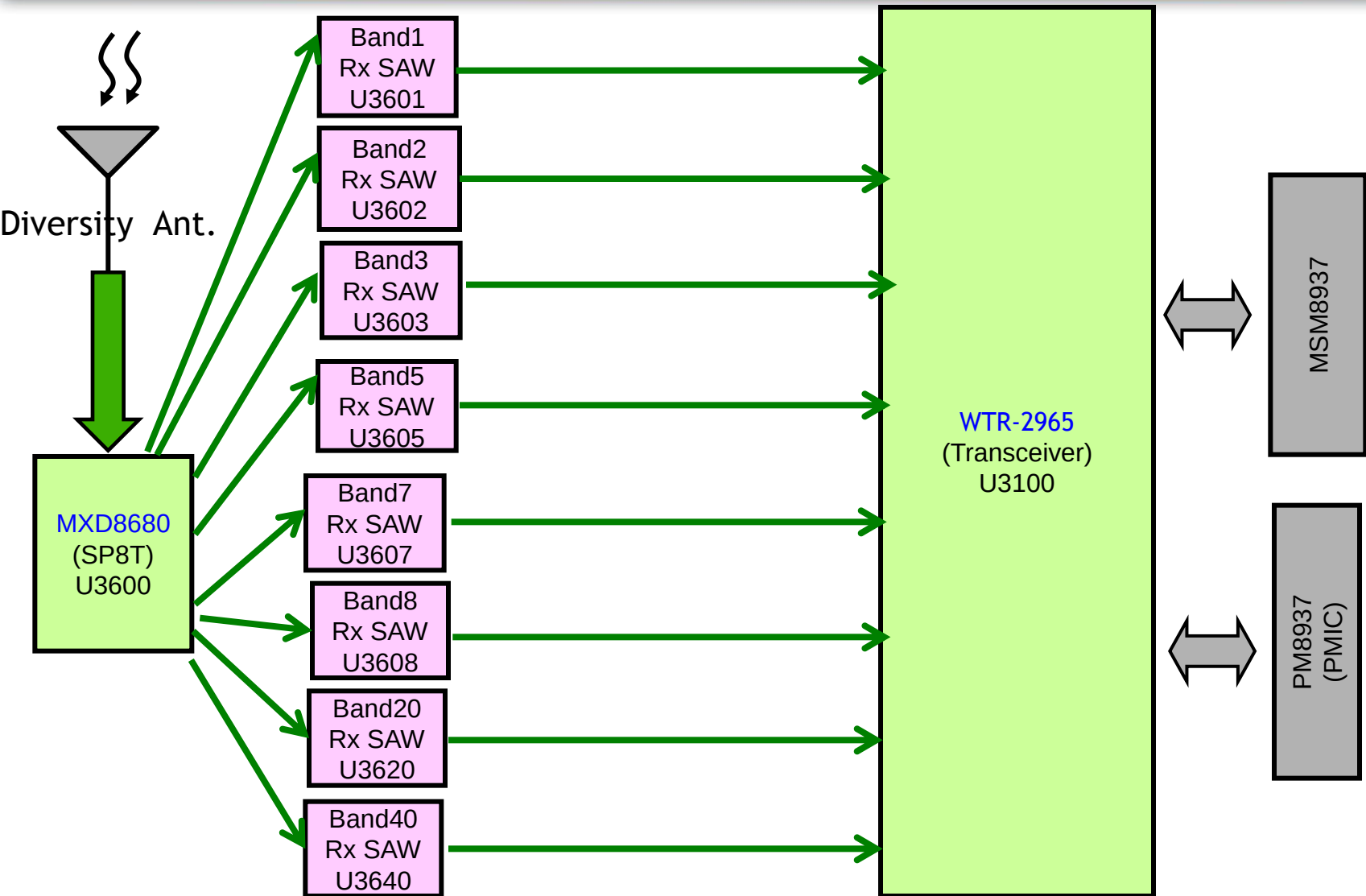
1. U3100: Transceiver
WTR-2965
2. U3201 : Band40 TX SAW
3. U3301: TXM
NZ5596E
4. U3202: MMBB
NZ5627E



Rx comp.
Tx comp.
Serial IC.
TRX

RF (Bottom)

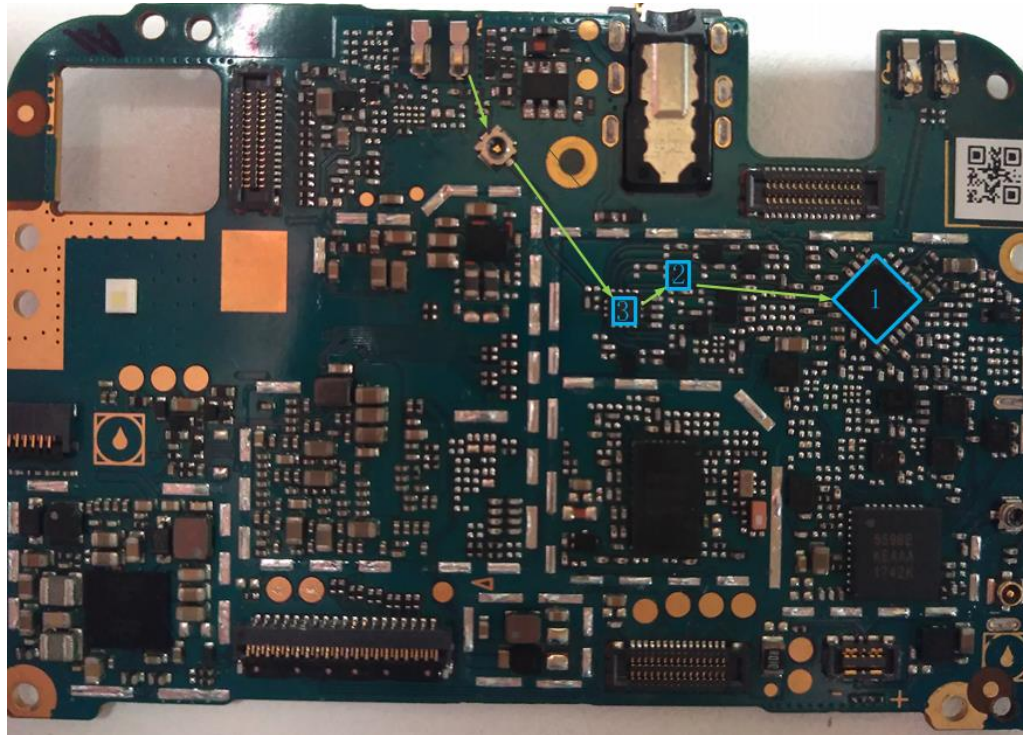
LTE Rx Diversity Ant Block Diagram



DRx Band 1

 Rx path

1. U3100 : Transceiver
WTR-2965
2. U3601: RX SAW
3. U3600: SP8T switch
MXD8680



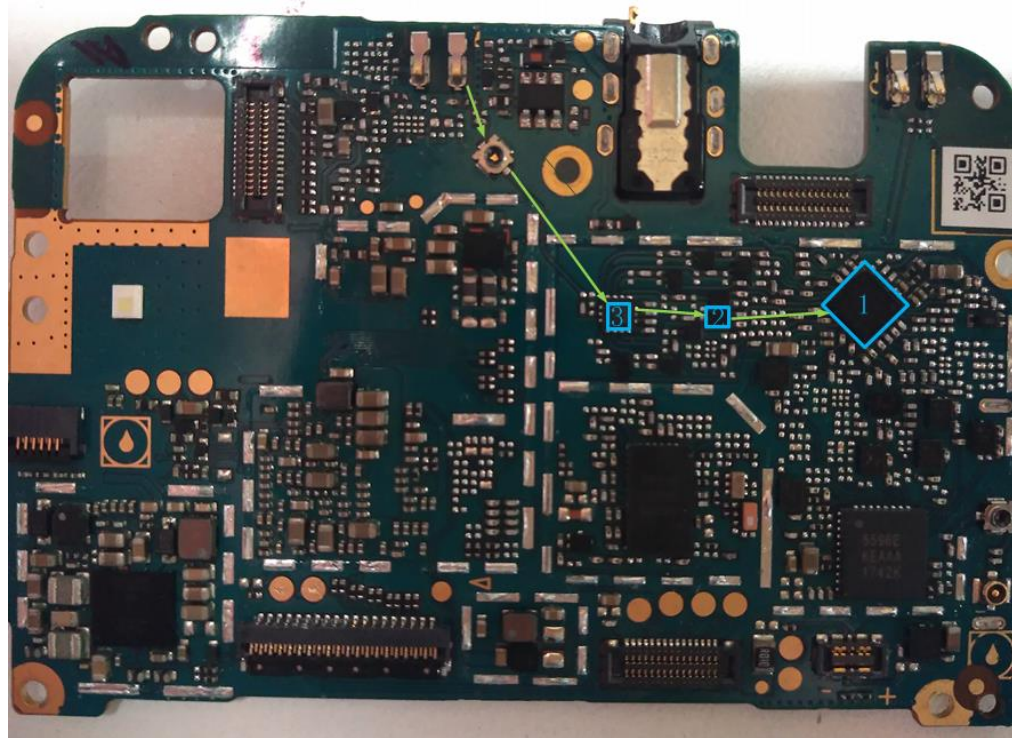
 Serial IC.
 Rx comp.

RF (Bottom)

DRx Band 2

 Rx path

1. U3100: Transceiver
WTR-2965
2. U3602: RX SAW
3. U3600 : SP8T switch
MXD8680



 Serial IC.

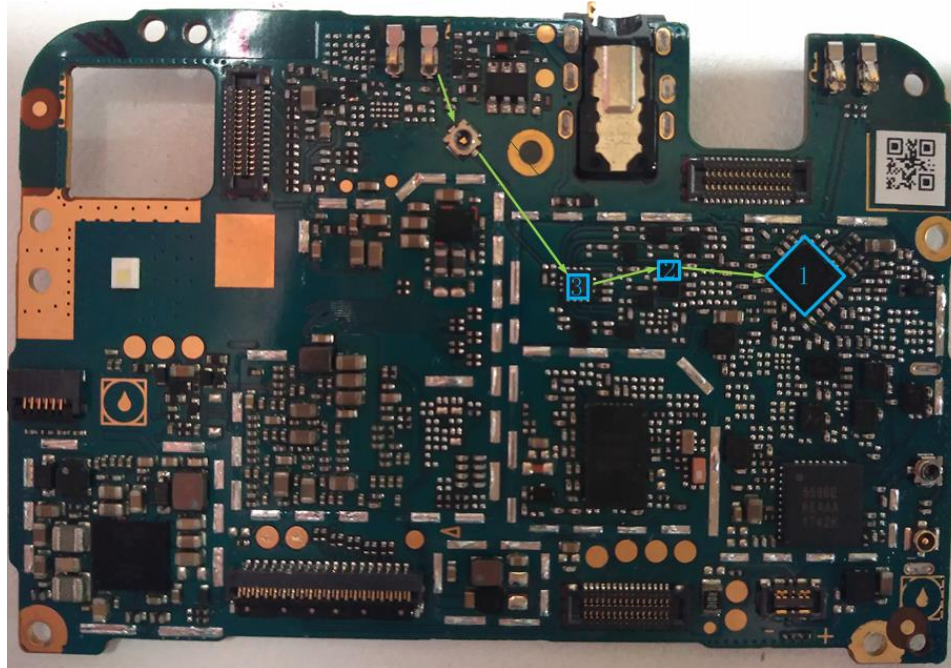
 Rx comp.

RF (Bottom)

DRx Band 3

 Rx path

1. U3100: Transceiver
WTR-2965
2. U3603: RX SAW
3. U3600: SP8T switch
MXD8680



 Serial IC.

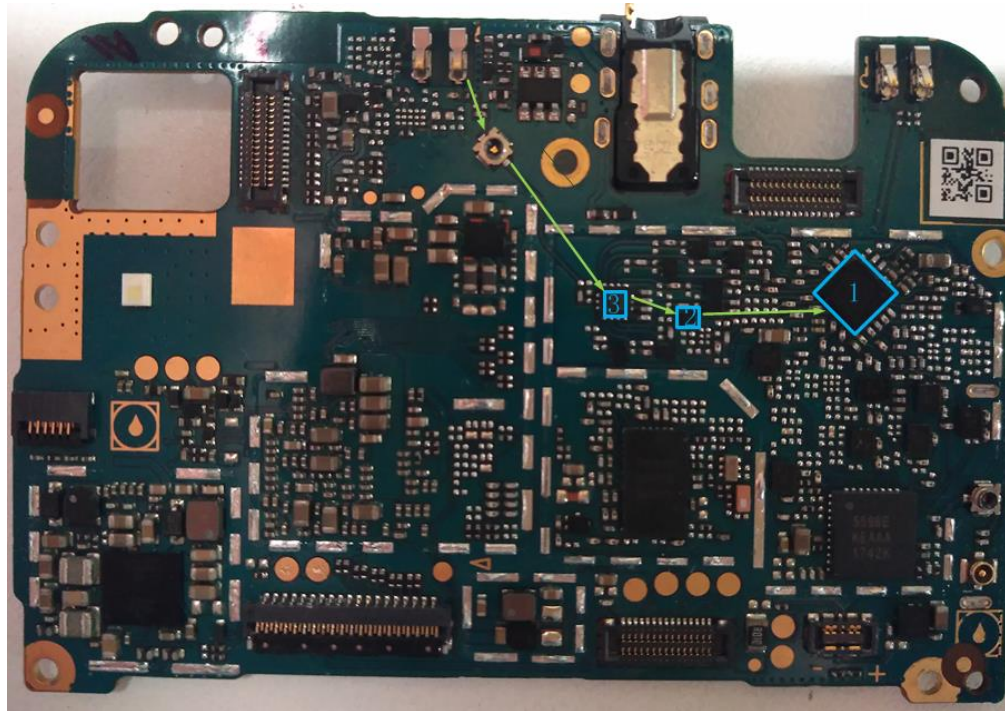
 Rx comp.

RF (Bottom)

DRx Band 5

 Rx path

1. U3100: Transceiver
WTR-2965
2. U3605: RX SAW
3. U3600: SP8T switch
SKY13418



 Serial IC.

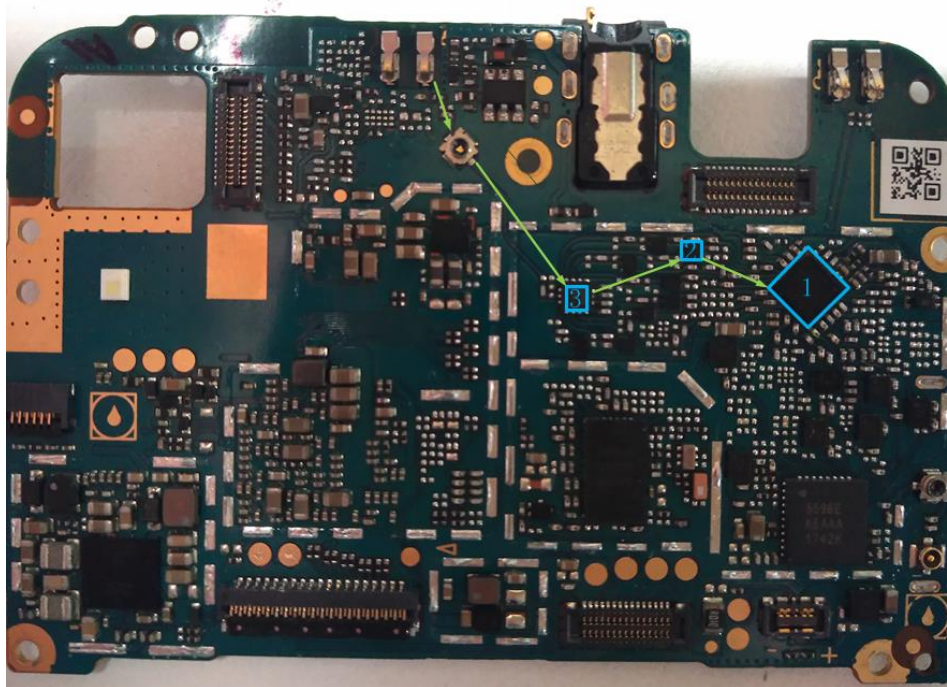
 Rx comp.

RF (Bottom)

DRx Band 7

 Rx path

1. U3100: Transceiver
WTR-2965
2. U3607: RX SAW
3. U3600: SP8T switch
MXD8680



 Serial IC.

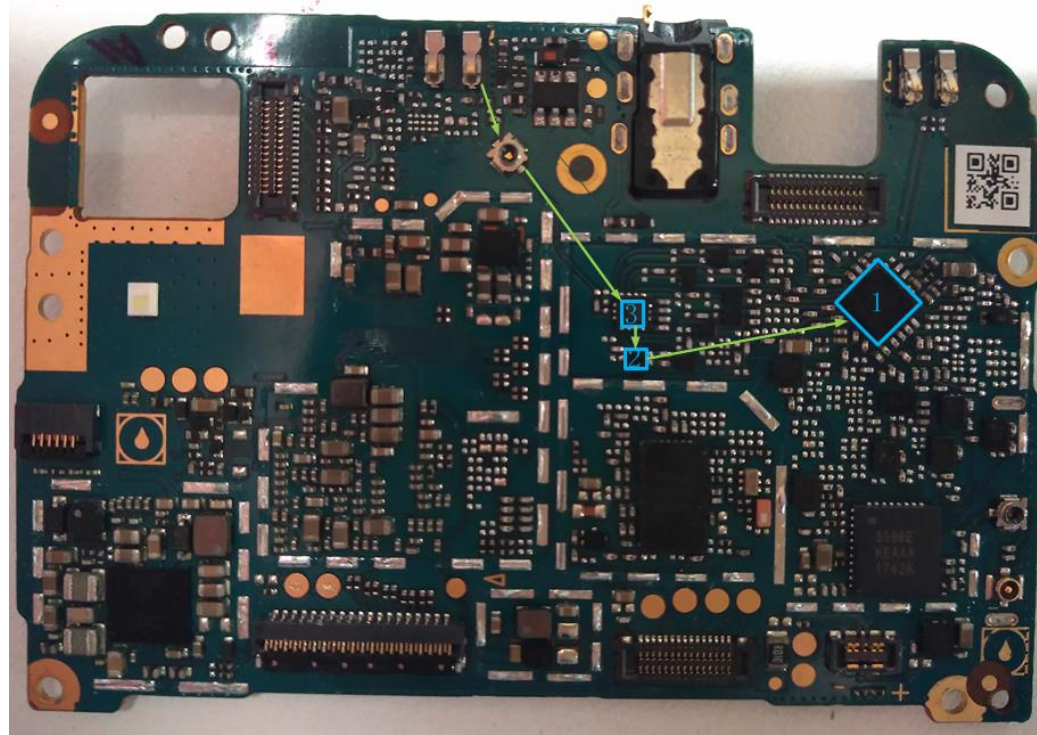
 Rx comp.

RF (Bottom)

DRx Band 8

 Rx path

1. U3100: Transceiver
WTR-2965
2. U3608:RX SAW
3. U3600: SP8T switch
MXD8680



 Serial IC.

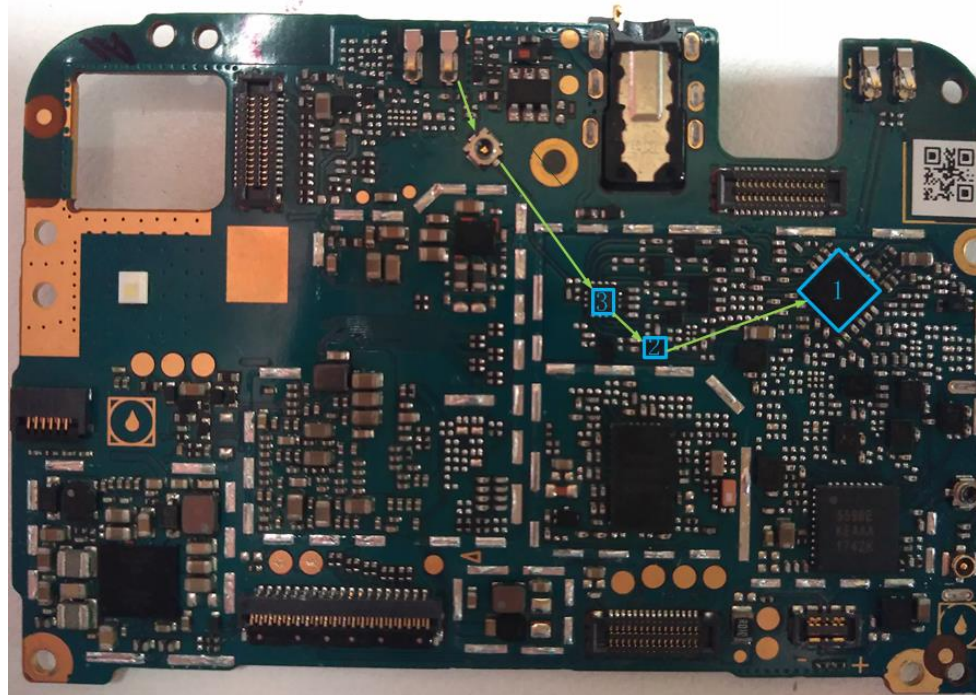
 Rx comp.

RF (Bottom)

DRx Band 20

 Rx path

1. U3100: Transceiver
WTR-2965
2. U3620:RX SAW
3. U3600: SP8T switch
MXD8680



 Serial IC.

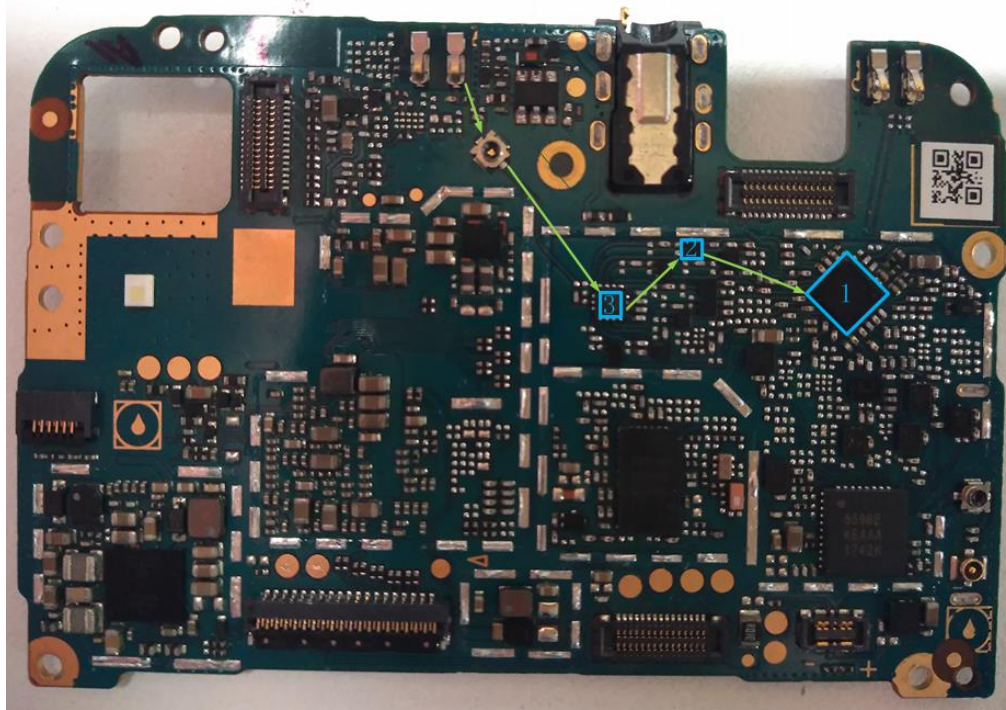
 Rx comp.

RF (Bottom)

DRx Band 40

 Rx path

1. U3100: Transceiver
WTR-2965
2. U3640:RX SAW
3. U3600: SP8T switch
MXD8680

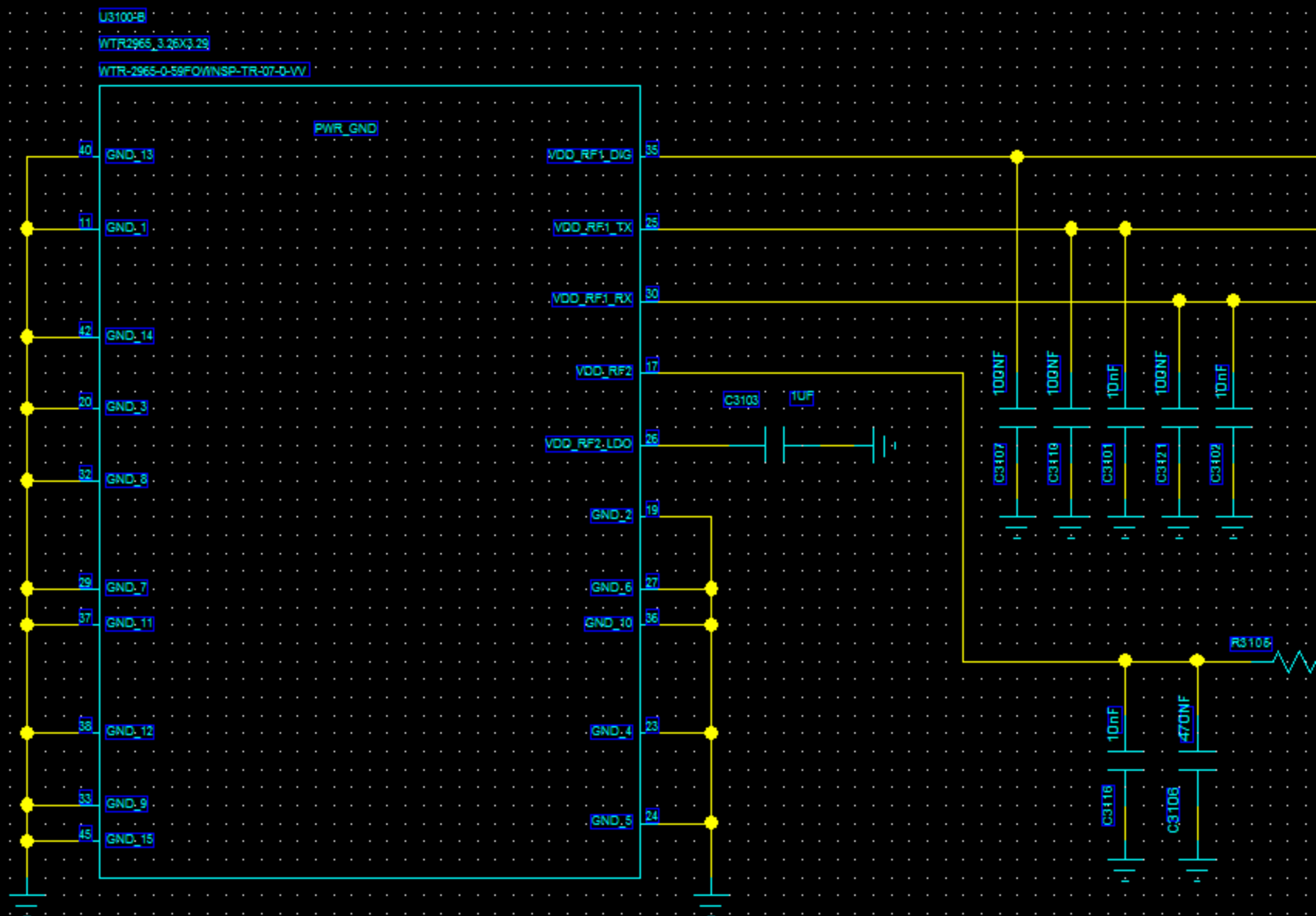


 Serial IC.

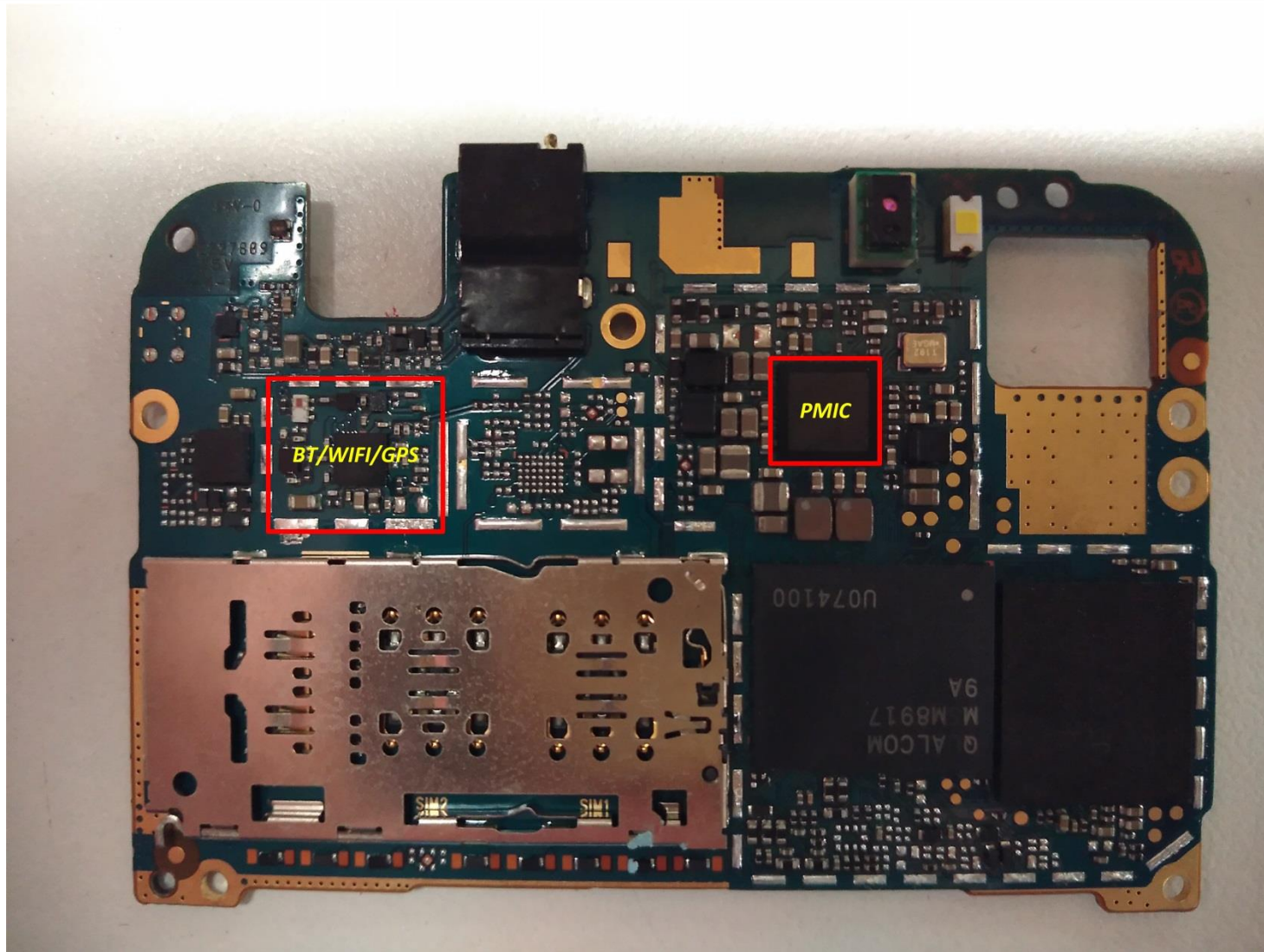
 Rx comp.

RF (Bottom)

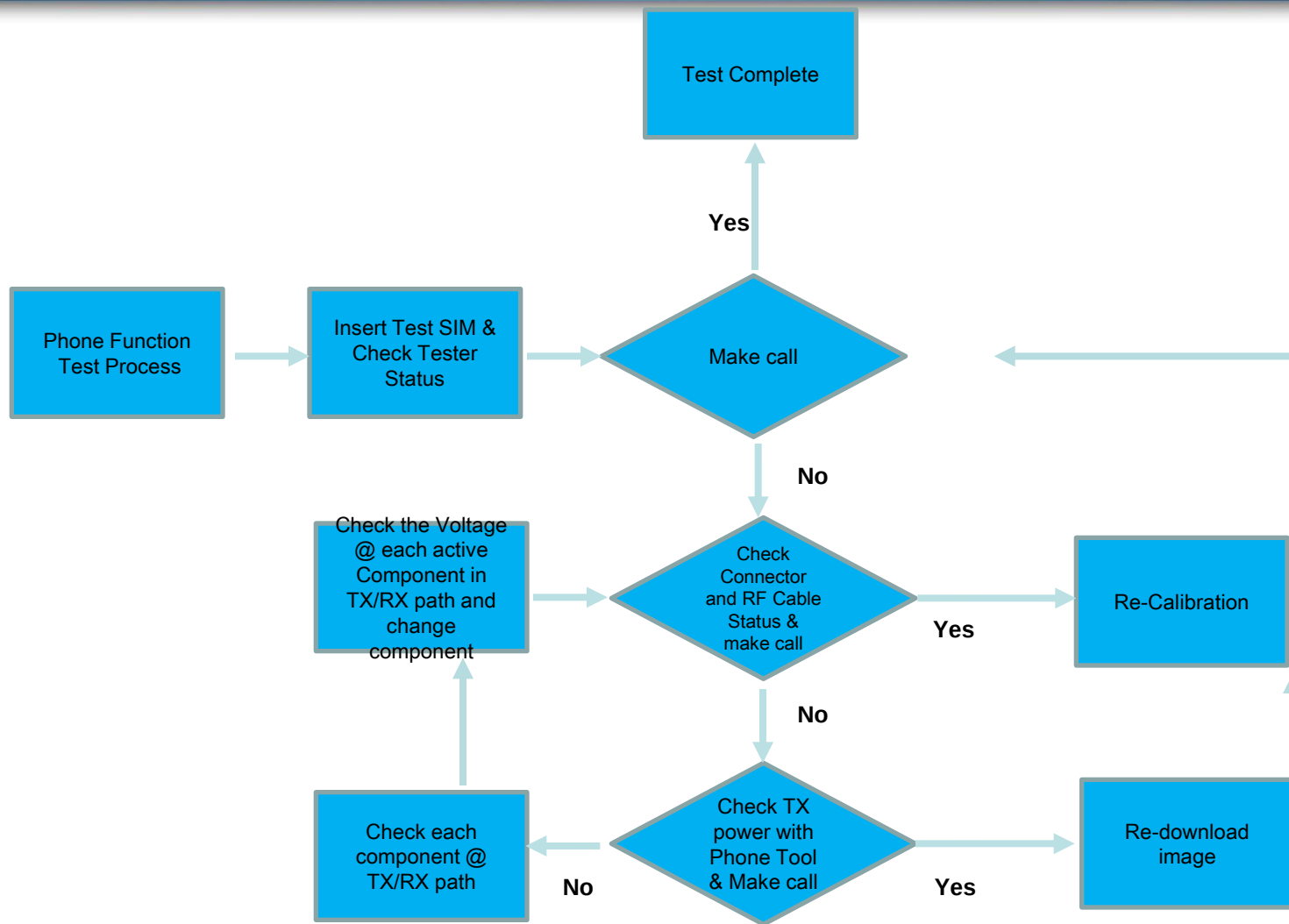
WTR2965 PWR & GND



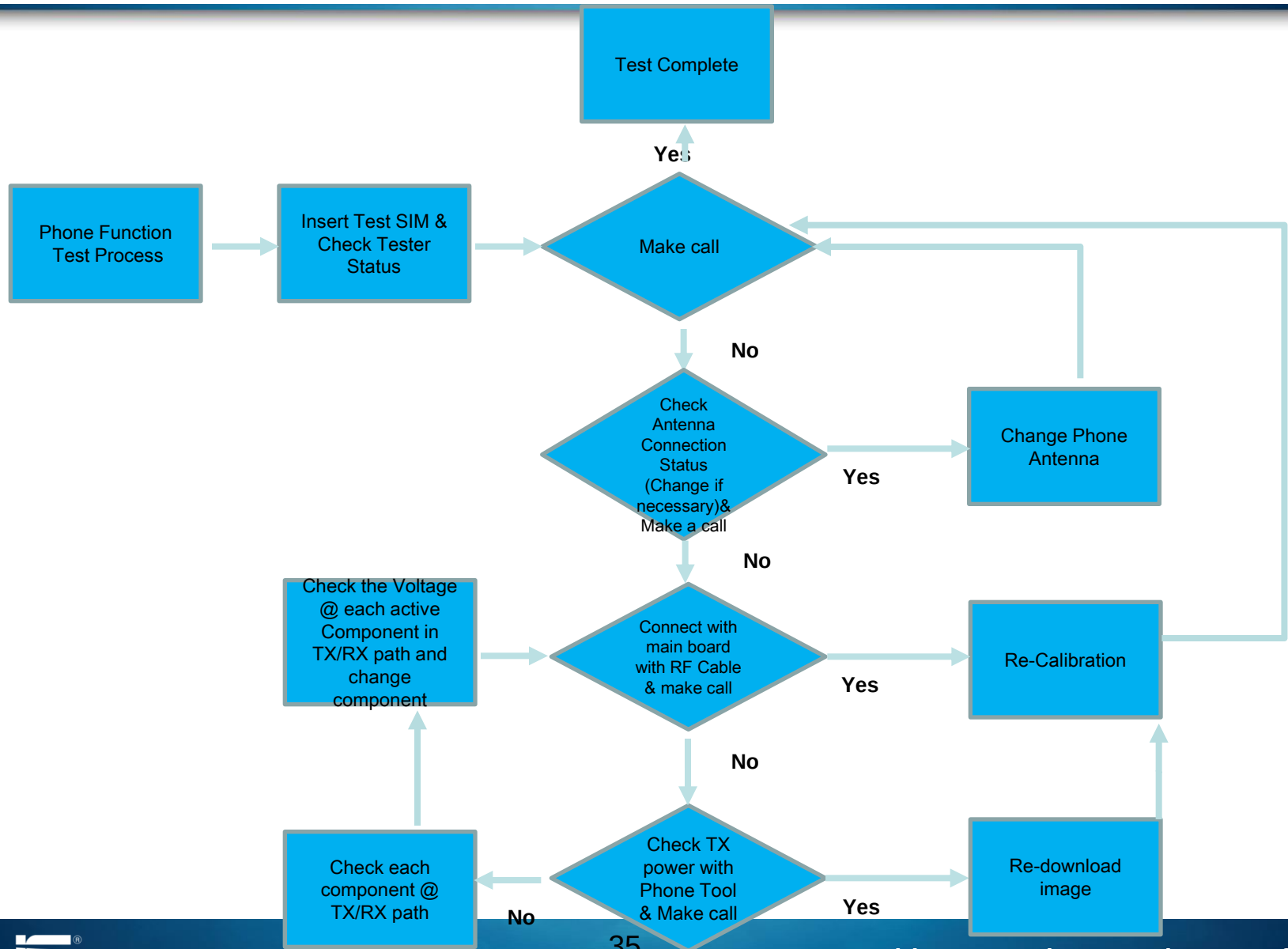
PMIC Location (Top)



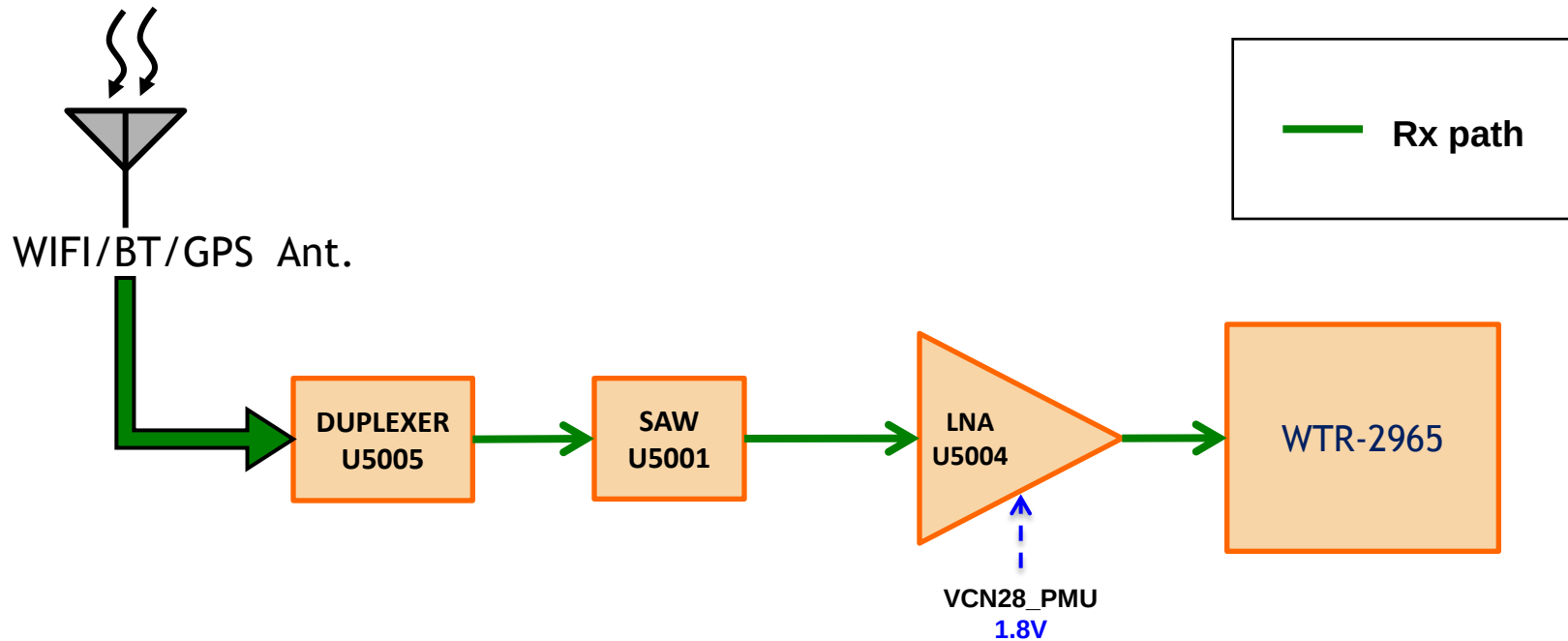
Board Level Phone Function Test Procedure



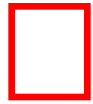
System Level Phone Function Test Procedure



GPS Block Diagram



GPS

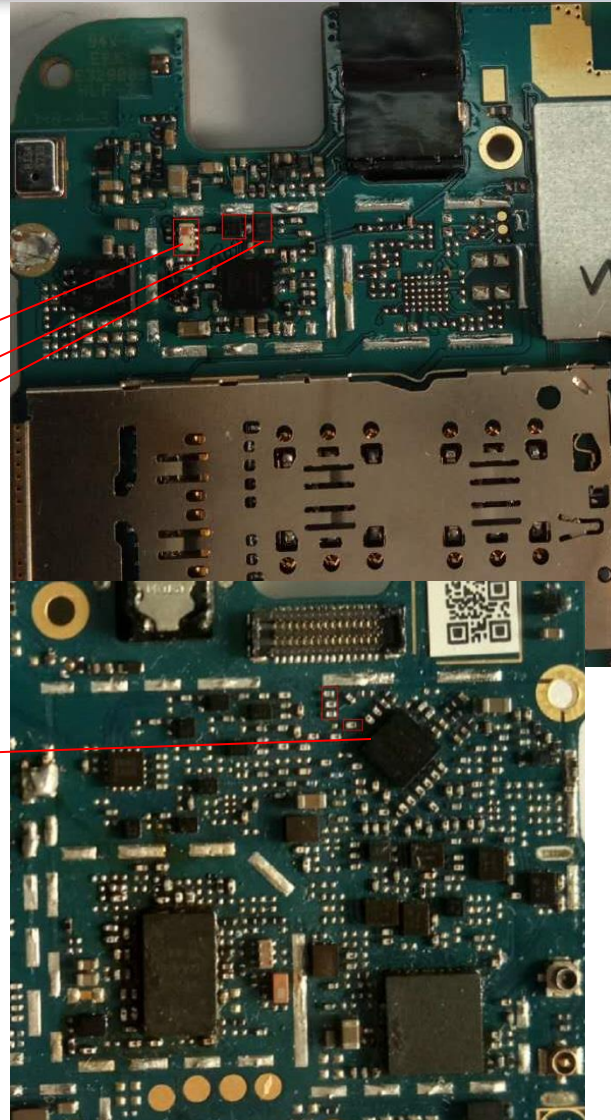


Rx comp.

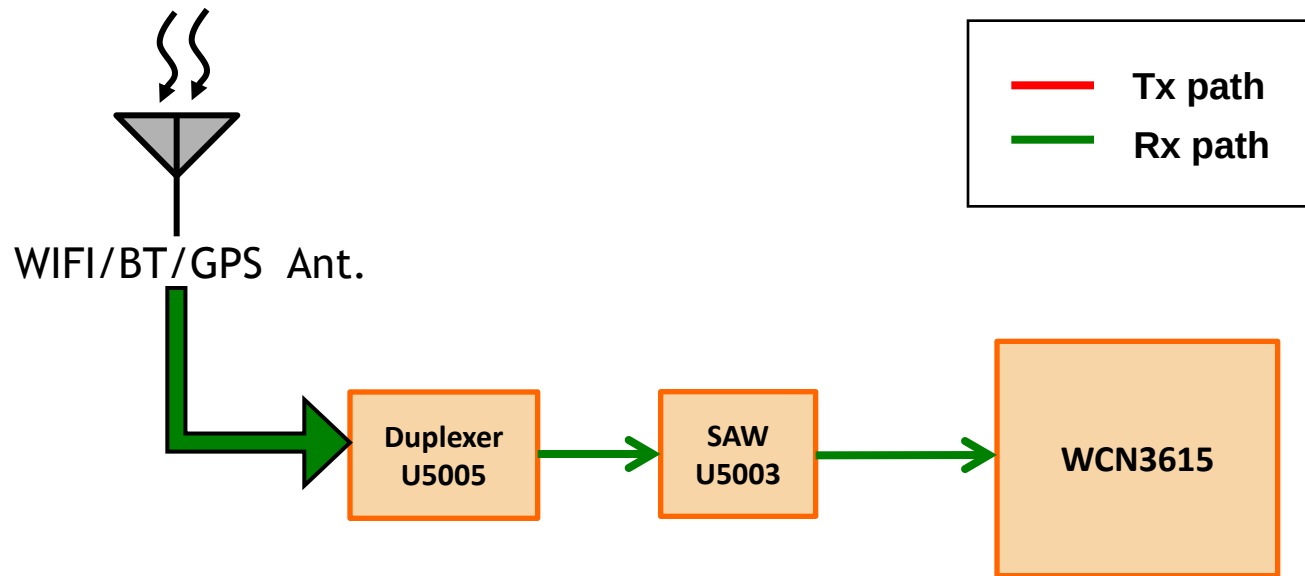


Serial IC.

- 1.U5004 WBG 双工
- 2.U5001 GPS SAW
- 3.U5005 GPS LAN
- 4. U3100 WTR2965

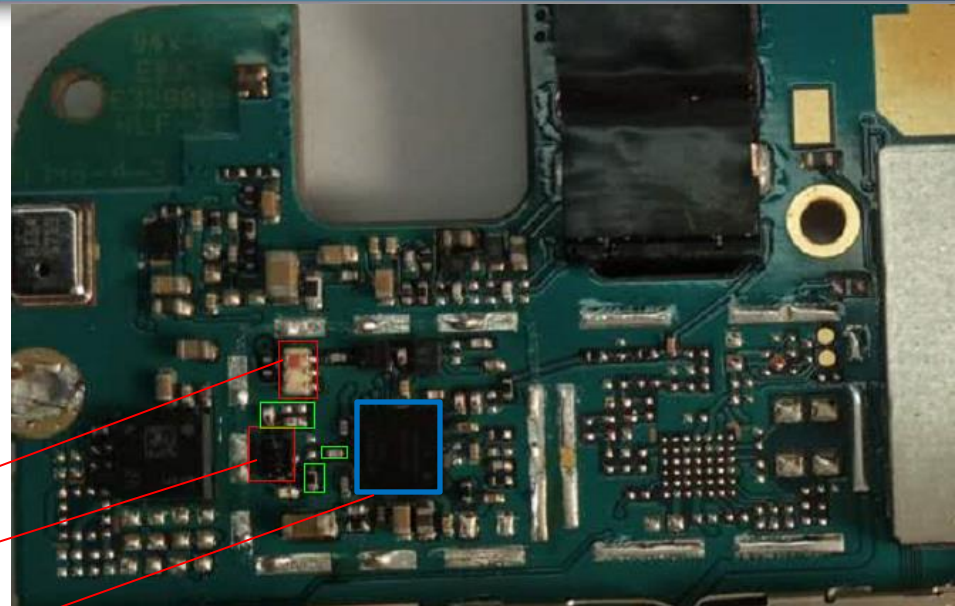


WIFI/BT Block Diagram



WIFI/BT 2.4G

-  Match net
-  TRx comp
-  Serial IC.



1. U5005: **Duplexer**
2. U5003: SAW
3. U5002: WIFI/BT Module
WCN3615

BT/WLAN Function Test Procedure

- **Change antenna**
- **Check the BT/WLAN feed point**
- **Check WCN3615(U5002) and surrounding components**

GPS Function Test Procedure

- **Check GPS function with GPS Test Tool.**
- **Check the C/N value with different signal level @ -80 dBm & -120 dBm.**
- **Check the GPS feed point.**
- **Check WTR2965(U3100) and surrounding components**
- **Check the 1.8V supply voltage of GPS LNA**

Thanks for Your attentions !